

Exam Date & Time: 17-Jul-2023 (09:15 AM - 12:30 PM)



CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

University Examination July-2023

B.Tech (CE)-VII

09:15 a.m. to 12:30 p.m.

Design of Language Processors [CE442]

Marks: 70

Duration: 195 mins.

Section-I

Answer all the questions.

Section Duration: 40 mins

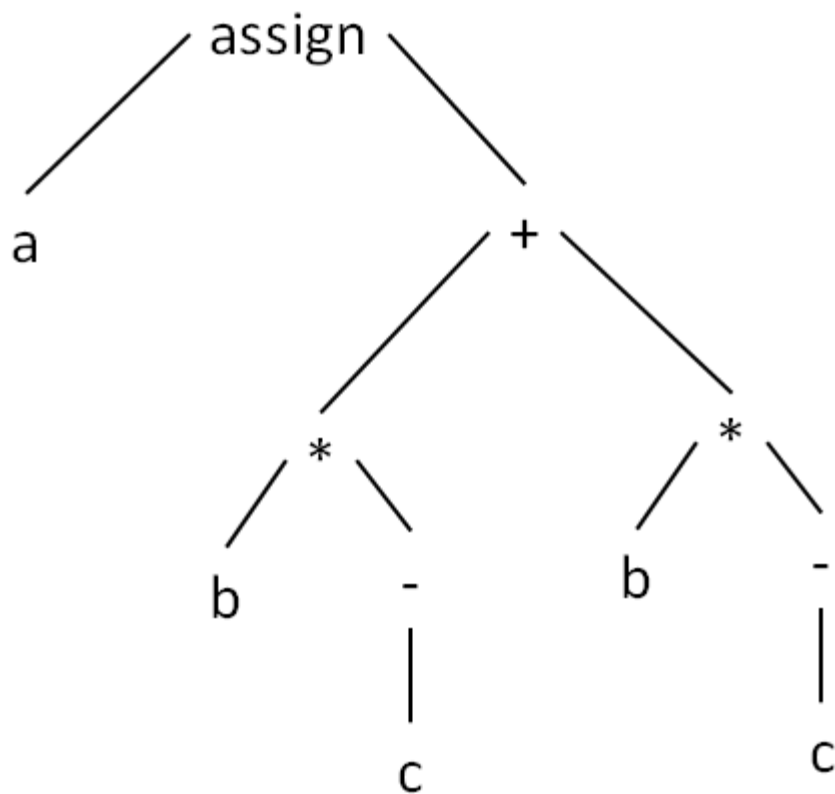
1			Which one of the following is not language processor?	(1)
			1) Preprocessor 2) Translator 3) Detranslator 4) Predector	
2			Analysis phase of compiler is also called ____	(1)
			1) Front End 2) Back End	
3			Gap between Application domain and Execution domain is called ____	(1)
			1) Semantic gap 2) Specification gap 3) Execution gap	
4			Which of the following is language traslator?	(1)
			1) Compiler 2) C Preprocessor 3) VS code	
5			In case of nested macro call, which macro is processed first?	(1)
			1) Inner macro 2) Outer macro	
6			Which one of the following table is not generated by two pass assembler?	(1)
			1) SYMTAB 2) POOLTAB 3) LITAB 4) TII	
7			Bootstrap loader is responsible to load OS.	(1)
			1) TRUE 2) FALSE	
8			Symbol table is created by which phase of compiler?	(1)
			1) Lexical Analyser 2) Code Optimizer 3) Intermediate Code Generator 4) Semantic Analyser	
9			Which of the following data structure is best for implementation of symbol table?	(1)
			1) Linear list 2) Hash Table 3) Matrix	
10			Fullform of SDD is ____	(1)
			1) Syntax Directed Definition 2) System Design Domain 3) Software Directed Definition	
11			Which of the following is not code optimization technique?	(1)
			1) Common subexpression elimination 2) Dead-code elimination 3) Reduction in strength 4) RDP	
12			Heap allocation is done at run time.	(1)
			1) TRUE 2) FALSE	

13			Binding of name with storage is called ____	(1)
			1) Environment 2) State 3) Value	
14			Which of the following string is not accepted by the $1(0 1)^*0$?	(1)
			1) 10 2) 1100 3) 1111 4) 1010	
15			Macro is ____ subroutine	(1)
			1) Close 2) Open	
16			Linking of the object modules are done by ____	(1)
			1) Linker 2) Loader 3) Compiler 4) Assembler	
17			RDP cannot handle the left recursive grammar.	(1)
			1) TRUE 2) FALSE	
18			LL(1) is top-down parsing method.	(1)
			1) TRUE 2) FALSE	
19			Separate symbol table is generated for each scope in the program.	(1)
			1) TRUE 2) FALSE	
20			Code optimization is the part of back end.	(1)
			1) TRUE 2) FALSE	

Section-II

Answer 6 out of 9 questions.

1			List out the different phases of compiler. Explain significance of each phase of compiler with an example.	(7)
2			Compute First and Follow sets for non-terminal in following grammar. $E \rightarrow TE'$ $E' \rightarrow +TE' \mid \wedge$ $T \rightarrow FT'$ $T' \rightarrow *FT' \mid \wedge$ $F \rightarrow id \mid (E)$	(7)
3			Compute the operator precedence matrix and precedence function for the following grammar. +, *, id are terminal symbols. $E \rightarrow E + T \mid T$ $T \rightarrow T * F \mid F$ $F \rightarrow id$	(7)
4			Construct NFA for $a^*b^*(c d)e$ using Thompson's Construction.	(7)
5			List issues in the design of code generator.	(7)
6			Generate the LR(0) item set for following grammar: $S \rightarrow aSb \mid ab$	(7)
7			Prepare Quadruple for the given syntax tree.	(7)



8			Why there is a need to have code optimization phase? Classify optimization types with an example.	(7)
9			Explain Error Recovery Strategies in detail with suitable example.	(7)

Section-III

Answer 2 out of 3 questions.

1			Compare variant-I and variant-II of intermediate code (IC).	(4)
2			Explain absolute loader.	(4)
3			Show that given grammar is ambiguous. $S \rightarrow ABA$ $A \rightarrow aA \mid \wedge$ $B \rightarrow bB \mid \wedge$	(4)

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